Chapter 1

Introduction to Graph Theory

Remark 1.1 (Chapter goals). The goal of this chapter is to introduce the basic concepts of graph theory and its vocabulary. We also discuss a little of its history.

1.1 Graphs, Multigraphs, and Simple Graphs

Definition 1.2 (Graph). A graph is a tuple $G = (V, E)$ where V is a (finite) set of vertices and E is a finite collection of edges. The set E contains elements from the union of the one- and two-element subsets of V. That is, each edge is either a one- or two-element subset of V.

Remark 1.3. It is generally easiest to visualize a graph as a collection of shapes for its vertices and lines (or curves) connecting the vertices for its edges. Though any shape can be used for the vertices, in mathematical graph theory, dots or circles are the most common.

Example 1.4. Consider the set of vertices $V = \{1, 2, 3, 4\}$ and the set of edges

$$E = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}\}.$$

Then, the graph $G = (V, E)$ has four vertices and four edges. See Fig. 1.1 for the visual representation of G.

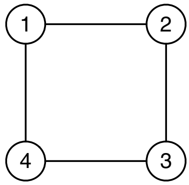

Fig. 1.1 It is easier for explanations to represent a graph by a diagram in which vertices are represented by points (or squares, circles, triangles, etc.) and edges are represented by lines connecting vertices.

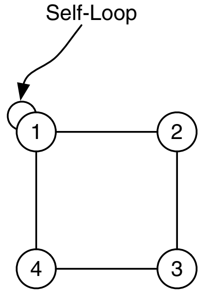

Fig. 1.2 A self-loop is an edge in a graph G that contains exactly one vertex. That is, an edge that is a one-element subset of the vertex set. Self-loops are illustrated by loops at the vertex in question.

Definition 1.5 (Self-loop). If $G = (V, E)$ is a graph and $v \in V$ and $e = \{v\}$, then edge e is called a *self-loop*. That is, any edge that is a single-element subset of V is called a self-loop.

Example 1.6. If we replace the edge set in Example 1.4 with

$$E = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}, \{1\}\},$$

then the visual representation of the graph includes a self-loop that starts and ends at Vertex 1. This is illustrated in Fig. 1.2.

Definition 1.7 (Vertex adjacency). Let $G = (V, E)$ be a graph. Two vertices v_1 and v_2 are said to be *adjacent* if there exists an edge $e \in E$ so that $e = \{v_1, v_2\}$. A vertex v is self-adjacent if $e = \{v\}$ is an element of E.

Definition 1.8 (Edge adjacency). Let $G = (V, E)$ be a graph. Two edges e_1 and e_2 are said to be *adjacent* if there exists a vertex

v so that v is an element of both e_1 and e_2 (as sets). An edge e is said to be *adjacent* to a vertex v if v is an element of e as a set.

Definition 1.9 (Neighborhood). Let $G = (V, E)$ be a graph, and let $v \in V$. The *neighbors* of v are the set of vertices that are adjacent to v. Formally,

$$N(v) = \{u \in V : \exists e \in E\,(e = \{u, v\} \text{ or } u = v \text{ and } e = \{v\})\}. \quad (1.1)$$

In some texts, $N(v)$ is called the *open neighborhood* of v, while $N[v] = N(v) \cup \{v\}$ is called the *closed neighborhood* of v. This notation is somewhat rare in practice. When v is an element of more than one graph, we write $N_G(v)$ as the neighborhood of v in graph G.

Remark 1.10. Equation (1.1) is read:

> $N(v)$ is the set of vertices u in (the set) V such that there exists an edge e in (the set) E so that $e = \{u, v\}$ or $u = v$ and $e = \{v\}$.

The logical expression $\exists x\,(R(x))$ is always read in this way; that is, there exists x so that some statement $R(x)$ holds. Similarly, the logical expression $\forall y\,(R(y))$ is read:

> For all y the statement $R(y)$ holds.

Admittedly, this sort of thing is very pedantic, but logical notation can help immensely in simplifying complex mathematical expressions.

Remark 1.11. The difference between the open and closed neighborhoods of a vertex can get a bit odd when you have a graph with self-loops. Since this is a highly specialized case, usually the author (of the paper, book, etc.) will specify a behavior.

Example 1.12. In the graph in Example 1.12, the neighborhood of Vertex 1 consists of Vertices 2 and 4 and Vertex 1 because Vertex 1 is adjacent to itself. The neighborhood of Vertex 1 now consists of vertices 1, 2, and 4. Note that the self-loop forces Vertex 1 into its own neighborhood.

Definition 1.13 (Degree). Let $G = (V, E)$ be a graph, and let $v \in V$. The *degree* of v, written $\deg(v)$, is the number of non-self-loop edges adjacent to v plus two times the number of self-loops defined at v. More formally,

$$\deg(v) = |\{e \in E : \exists u \in V(e = \{u, v\})\}| + 2\,|\{e \in E : e = \{v\}\}|\,.$$

Here, if S is a set, then $|S|$ is the cardinality of that set.

Remark 1.14. Note that each vertex in the graph in Fig. 1.1 has a degree of 2.

Example 1.15. Consider the graph shown in Example 1.6. The degree of Vertex 1 is 4. We obtain this by counting the number of non-self-loop edges adjacent to Vertex 1 (there are 2) and adding two times the number of self-loops at Vertex 1 (there is 1) to obtain $2 + 2 \times 1 = 4$.

Remark 1.16. Let $G = (V, E)$ be a graph. There are two degree values that are of interest in graph theory: The largest and smallest vertex degrees are usually denoted by $\Delta(G)$ and $\delta(G)$, respectively. That is,

$$\Delta(G) = \max_{v \in V} \deg(v) \quad \text{and} \tag{1.2}$$

$$\delta(G) = \min_{v \in V} \deg(v). \tag{1.3}$$

Definition 1.17 (Multigraph). A graph $G = (V, E)$ is a *multigraph* if there are two edges e_1 and e_2 in E so that e_1 and e_2 are equal as sets. That is, there are two vertices v_1 and v_2 in V so that $e_1 = e_2 = \{v_1, v_2\}$.

Remark 1.18. Note in the definition of graph (Definition 1.2), we were very careful to specify that E is a *collection* of one- and two-element subsets of V rather than to say that E was a set. This allows us to have duplicate edges in the edge set and thus to define multigraphs. In computer science, a set that may have duplicate entries is sometimes called a *multiset*. A multigraph is a graph in which E is a multiset.

Derivation 1.19 (Königsburg bridge problem). Graph theory began with Leonhard Euler with his study of the bridges of Königsburg problem. Here's how it started: The city of Königsburg exists as a collection of islands connected by bridges. The problem

Euler wanted to analyze was: Is it possible to go from island to island, traversing each bridge only once?

Following Euler, we construct a graph to analyze the bridges of Königsburg problem. Assume that we treat each island as a vertex and each bridge as an edge. The resulting multigraph is illustrated in Fig. 1.3. The edge collection is

$$E = \{\{A, B\}, \{A, B\}, \{A, C\}, \{A, C\}, \{A, D\}, \{B, D\}, \{C, D\}\}.$$

This multigraph occurs because there are two bridges connecting island A with island B and two bridges connecting island A with island C. If two vertices are connected by two (or more) edges, then the edges are simply represented as parallel lines (or arcs) connecting the vertices.

Note that this representation dramatically simplifies the analysis of the problem in so far as we can now focus only on the structural properties of this graph. It's easy to see (from Fig. 1.3) that each vertex has an odd degree. More importantly, since we are trying to traverse islands without ever recrossing the same bridge (edge), when we enter an island (say C), we use one of the three edges. Unless this is our final destination, we must use *another* edge to leave C. Additionally, assuming we have not crossed all the bridges yet, we know that we *must* leave C. That means that the third edge that

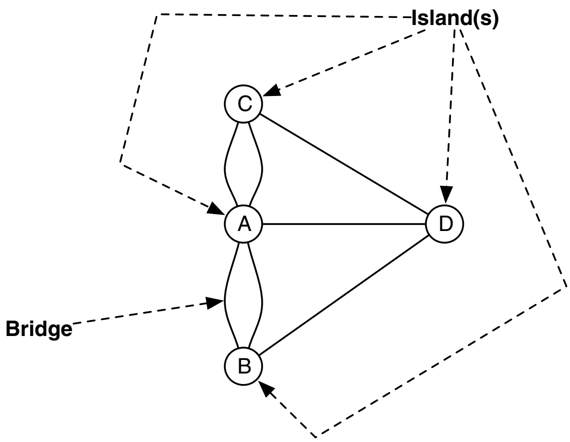

Fig. 1.3 Representing each island as a dot and each bridge as a line or curve connecting the dots simplifies the visual representation of the seven Königsburg bridges.